Glossary

1/f noise: A type of random noise that increases in amplitude at lower frequencies. It is widely observable in physical systems, but not well understood. See *white noise* for comparison.

-3dB cutoff frequency: The division between a filter's passband and transition band. Defined as the frequency where the frequency response is reduced to -3dB (0.707 in amplitude).

"A" law: Companding standard used in Europe. Allows digital voice signals to be represented with only 8 bits instead of 12 bits by making the quantization levels unequal. See *mu law* for comparison.

AC: Alternating Current. Electrical term for the portion of a signal that fluctuates around the average (DC) value.

Accuracy: The error in a measurement (or a prediction) that is repeatable from trial to trial. Accuracy is limited by systematic (repeatable) errors. See *precision* for comparison.

Additivity: A mathematical property that is necessary for linear systems. If input a produces output p, and if input b produces output q, then an input of $a+b$ produces an output of $p+q$.

Aliasing: The process where a sinusoid changes from one frequency to another as a result of sampling or other nonlinear action. Usually results in a loss of the signal's information.

Amplitude modulation: Method used in radio communication for combining an information carrying signal (such as audio) with a carrier wave. Usually carried out by multiplying the two signals.

Analysis: The forward Fourier transform; calculating the frequency domain from the time domain. See *synthesis* for comparison.

Antialias filter: Low-pass analog filter placed before an analog-to-digital converter. Removes frequencies above one-half the sampling rate that would alias during conversion.

ASCII: A method of representing letters and numbers in binary form. Each character is assigned a number between 0 and 127. Very widely used in computers and communication.

Aspect ratio: The ratio of an image's width to its height. Standard television has an aspect ratio of 4:3, while motion pictures have an aspect ratio of 16:9.

Assembly: Low-level programming language that directly manipulates the registers and internal hardware of a microprocessor. See *high-level language* for comparison.

Associative property of convolution: Written as: $(a[n] * b[n]) * c[n] = a[n] * (b[n] * c[n])$. This is important in signal processing because it describes how cascaded stages behave.

Autocorrelation: A signal correlated with itself. Useful because the Fourier transform of the autocorrelation is the power spectrum of the original signal.

Backprojection: A technique used in computed tomography for reconstructing an image from its views. Results in poor image quality unless used with a more advanced method.

BASIC: A high-level programming language known for its simplicity, but also for its many weaknesses. Most of the programs in this book are in BASIC.

Basilar membrane: Small organ in the ear that acts as a spectrum analyzer. It allows different fibers in the cochlear nerve to be stimulated by different frequencies.

Basis functions: The set of waveforms that a decomposition uses. For instance, the basis functions for the Fourier decomposition are unity amplitude sine and cosine waves.